Group No.:5

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Context Handling For Follow-ups

Introduction:

In modern educational institutions, students frequently require information related to examinations, course schedules, academic calendars, and institutional policies. Traditionally, such information is provided through notice boards, faculty members, administrative offices, or official websites. However, these methods often lead to delays, confusion, and repetitive queries.

With the increasing adoption of artificial intelligence technologies, conversational agents or **chatbots** have emerged as an efficient solution to automate student support services. Chatbots allow users to interact with systems using natural language and receive instant responses without human intervention.

This project presents the development of an **Institute FAQ Chatbot with Multi-Turn Conversation Support**, designed to answer student queries related to examination schedules. The chatbot utilizes **Natural Language Processing (NLP)** techniques to understand user input, extract relevant entities such as **course name and semester**, and generate appropriate responses.

Unlike traditional chatbots that respond to single queries only, this system supports **multi-turn conversations**. This means the chatbot can remember previous interactions and combine them with current queries to provide meaningful responses.

For example:

User: *When is the exam?*
Bot: *Which course and semester?*
User: *Third year CS*

The chatbot intelligently combines both inputs to answer the question accurately.

This project demonstrates how basic NLP techniques and contextual memory can be used to build an efficient conversational system for educational institutions.

Problem Statement:

Students often need quick access to exam schedules and academic information. However, traditional methods for accessing such information present several challenges:

1. Repetitive Queries

Administrative staff frequently receive the same questions from multiple students regarding exam dates, course schedules, or semester details.

2. Limited Availability

Students can only obtain information during office hours if they rely on faculty or administrative staff.

3. Poor User Experience

Students often need to search through lengthy documents or websites to find relevant information.

4. Inefficient Communication

Communication through emails or notice boards may lead to delays or misunderstandings.

5. Lack of Conversational Systems

Most existing FAQ systems require users to search manually and do not support conversational interaction.

Furthermore, many basic chatbots cannot handle **follow-up questions**. If a user asks a second question that depends on the previous one, the system fails to understand the context.

Example problem:

User:
"When is the exam?"

Bot:
"Please specify the course."

User:
"Third year."

Traditional systems treat this as a new query rather than continuing the previous conversation.

Therefore, there is a need to develop a chatbot system that:

- Understands natural language queries
- Extracts relevant information automatically
- Maintains conversation context
- Provides accurate responses quickly

Objectives

The primary objectives of this project are:

1. To design and develop an **automated chatbot system** capable of answering student queries related to exam schedules.
2. To implement **entity recognition** for extracting important information such as course and semester.
3. To support **multi-turn conversations** by maintaining context between user queries.

4. To reduce the workload of administrative staff by automating repetitive queries.
5. To improve accessibility by allowing students to receive information instantly.
6. To demonstrate the practical application of **Natural Language Processing techniques in educational systems**.

Scope of the Project:

The scope of this project includes the design and implementation of a chatbot system capable of handling exam-related queries within an academic institution.

The chatbot will:

- Accept queries from students
- Identify relevant information from the query
- Maintain conversational context
- Provide accurate responses using a predefined dataset

The system can be extended in the future to support:

- Course registration queries
- Attendance information
- Fee payment details
- Academic announcements

Although the current system focuses on exam schedules, the architecture allows future expansion to support a wider range of academic services.

Literature Review:

Chatbots have been widely used across industries such as customer service, healthcare, and education. Many educational institutions are adopting AI-based assistants to improve student engagement and support services.

Early chatbot systems were based on **rule-based approaches**, where predefined patterns were used to generate responses. One of the earliest examples is **ELIZA**, developed in the 1960s, which simulated conversations using pattern matching techniques.

With the advancement of artificial intelligence, modern chatbots incorporate **Natural Language Processing (NLP)** and **Machine Learning algorithms** to understand user intent and provide more intelligent responses.

In educational environments, chatbots have been used for:

- Student admission assistance
- Course information

- Assignment reminders
- Academic counseling

However, many chatbot systems still struggle with **multi-turn conversations**, where the system must remember previous interactions.

This project addresses that limitation by implementing a **conversation context manager**, allowing the chatbot to maintain relevant information across multiple messages.

Methodology:

The chatbot system follows a structured methodology consisting of multiple processing stages.

Step 1: User Query Input

The user enters a question related to exam schedules.

Example queries include:

- "When is the CS semester 5 exam?"
- "When is the exam?"
- "Third year IT"

The chatbot receives this input through a command-line interface or web interface.

Step 2: Text Preprocessing

The system processes the input text by:

- Converting text to lowercase
- Removing unnecessary punctuation
- Splitting sentences into tokens

This step ensures consistent processing of user queries.

Step 3: Entity Extraction

The chatbot identifies important entities from the query using pattern matching.

Entities include:

Entity	Example
Course	CS, IT
Semester	3, 5
Year	First year, Third year

Example:

Input query:

"CS semester 5 exam date"

Extracted entities:

Course → CS
Semester → 5

Step 4: Context Management

The system stores extracted information in a **conversation context object**.

The context contains:

- Current course
- Current semester
- Conversation history
- Pending information

This allows the chatbot to combine information across multiple messages.

Step 5: Database Query

Once the necessary information is available, the chatbot searches the **exam schedule database**.

Example database entry:

Course	Semester	Exam Date
CS	3	10 May 2026
CS	5	15 May 2026
IT	3	12 May 2026

Step 6: Response Generation

The chatbot constructs a response using the retrieved data.

Example:

"The CS Semester 5 exam is scheduled on 15 May 2026."

Step 7: Clarification Handling

If required information is missing, the chatbot asks a follow-up question.

Example:

User:
"When is the exam?"

Bot:
"Which course and semester?"

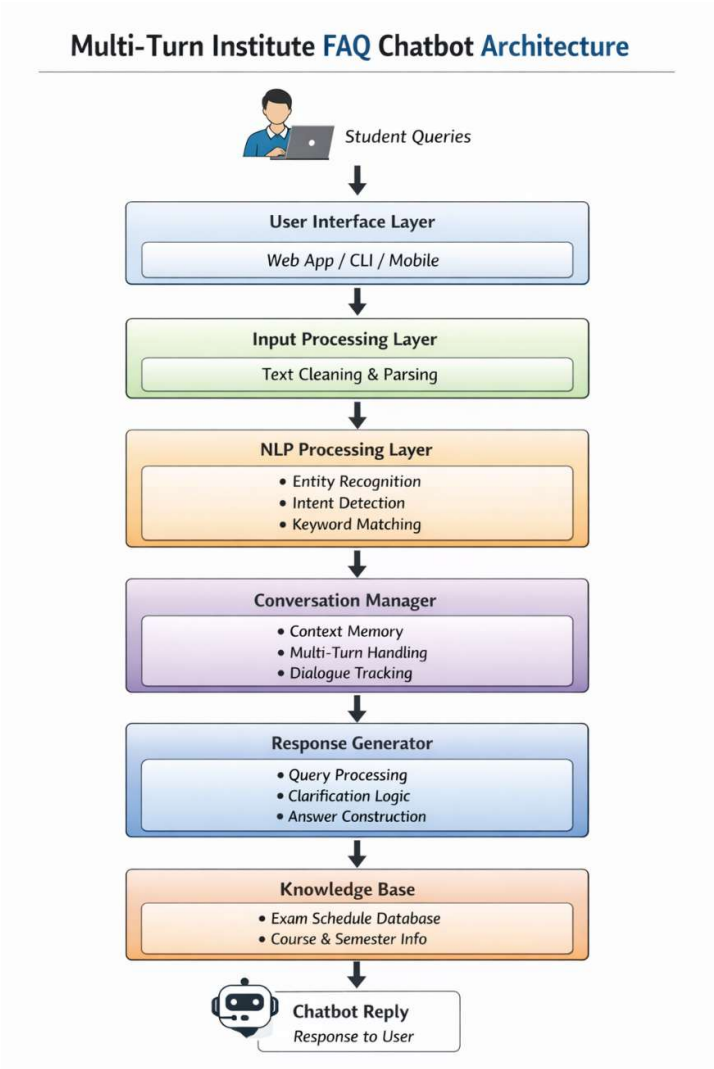
System Architecture:

The chatbot architecture consists of several layers that work together to process queries and generate responses.

Architecture Flow

User → Chat Interface → NLP Processing → Context Manager → Response Generator → Database → User Response

Each layer performs a specific function in the chatbot system.



Algorithm

The chatbot follows the following algorithm:

1. Start chatbot
2. Initialize conversation context
3. Accept user query
4. Preprocess input text
5. Extract entities
6. Update conversation context
7. Check if required information is available
8. If missing → ask clarification
9. If available → search database
10. Generate response
11. Display response
12. Wait for next query
13. Exit if user types "exit"

Advantages of the System

The proposed chatbot offers several benefits.

1. Instant Response

Students receive answers immediately without waiting for administrative assistance.

2. Reduced Workload

Reduces repetitive queries handled by staff.

3. 24/7 Availability

Students can access information anytime.

4. Improved User Experience

Students interact with a conversational interface.

5. Scalability

The system can be expanded to include more academic services.

Limitations

Despite its advantages, the system has certain limitations.

1. The chatbot uses rule-based NLP rather than advanced machine learning models.
2. It supports only predefined queries and entities.
3. The knowledge base is limited to exam schedules.
4. Complex natural language queries may not be fully understood.

Future Enhancements

Several improvements can be implemented in the future.

1. Integration with Machine Learning

Advanced NLP models such as BERT or GPT can improve understanding.

2. Voice-Enabled Chatbot

Voice assistants can allow students to interact through speech.

3. Web Deployment

The chatbot can be integrated into college websites.

4. Mobile Application

A dedicated mobile chatbot app can improve accessibility.

5. Multi-Language Support

The chatbot can support regional languages.

6. Real Database Integration

Integration with university databases for real-time updates.

Output Screenshots:

```
        print(f"Bot: {response}\n")

chatbot()
```

```
==================================================
  🎓  Exam Query Chatbot  (multi-turn enabled)
==================================================
  Ask about exam dates, e.g.:
    • 'When is the SEM 5 CS exam?'
    • 'When is the exam?'  →  'For third year CS'
  Type 'reset' to start over, 'exit' to quit.
==================================================

You:  3 sem exam?
Bot: Got it — Semester 3. Which department are you asking about? (e.g. CS, IT, ME, CE, EE)

You:  me
Bot: Sorry, I don't have the schedule for ME Semester 3 yet. Please check the official notice board.

You:  cs
Bot: The CS Semester 3 exam is scheduled on 10 May 2026. Best of luck! 🎓

You: ↑↓ for history. Search history with c-↑/c-↓
```

```
[ ]:
```

Conclusion:

This project demonstrates the development of an **Institute FAQ Chatbot with Multi-Turn Conversation Support** using basic Natural Language Processing techniques.

The chatbot can understand exam-related queries, extract important information such as course and semester, and provide accurate responses. By maintaining conversation context, the system supports multi-turn interactions, making it more natural and effective than traditional rule-based chatbots.

The project highlights the potential of conversational AI in improving communication within educational institutions. With further enhancements such as machine learning integration and web deployment, the chatbot can evolve into a comprehensive **AI-powered campus assistant** capable of handling a wide range of student queries.